

LABORATORY PRACTICE-II

CLOUD COMPUTING

Complete Q&A; for Final Oral Exam

Savitribai Phule Pune University | Third Year Computer Engineering (2019 Course)

Legend	Meaning
■ IMPORTANT	Very likely to be asked – prepare thoroughly
■ TRICKY	Conceptual/confusing questions examiners love
Q:	Standard question

SECTION 1: CLOUD COMPUTING BASICS

■ [IMPORTANT] What is Cloud Computing? Give an example.

Ans: Cloud computing means using the internet to access and use computing resources like servers, storage, databases, and software — instead of owning physical hardware. You pay only for what you use.

Ans: Example: Google Drive (storage), Gmail (email), Netflix (software) — all run on the cloud.

Q: Who is the father of Cloud Computing?

Ans: J.C.R. Licklider is considered the father of cloud computing. He introduced the idea of an 'Intergalactic Computer Network' in the 1960s. Joseph Carl Robnett Licklider is the full name.

■ [IMPORTANT] List the basic characteristics of Cloud Computing.

- On-demand self-service – Get resources anytime without human interaction
- Broad network access – Access from anywhere via internet
- Resource pooling – Resources are shared among multiple users
- Rapid elasticity – Scale up/down as needed quickly
- Measured service – Pay only for what you use (like electricity bill)

■ [IMPORTANT] What is Vertical and Horizontal Scaling?

Ans: Vertical Scaling (Scale Up): Increasing the power of an existing machine — add more RAM, CPU to the same server.

Ans: Horizontal Scaling (Scale Out): Adding more machines to distribute the load. E.g., adding more servers.

Note: Horizontal scaling is preferred in cloud as it's cheaper and has no downtime.

■ [IMPORTANT] What are the services provided by Cloud Computing?

Service	Full Form	Description	Example
IaaS	Infrastructure as a Service	Virtual machines, storage, network	AWS EC2, Azure VM
PaaS	Platform as a Service	Dev tools, runtime, database	Google App Engine
SaaS	Software as a Service	Ready-to-use software over internet	Gmail, Salesforce

■ [IMPORTANT] What are the deployment models in Cloud Computing?

- Public Cloud – Services open to everyone over internet (e.g., AWS, Azure, GCP)
- Private Cloud – Used exclusively by one organization (more secure)
- Hybrid Cloud – Combination of public and private cloud
- Community Cloud – Shared by organizations with common goals (e.g., government agencies)

Q: Difference between Cloud and Traditional Datacenters?

Cloud	Traditional Datacenter
Pay-as-you-go pricing	High upfront hardware cost
Easily scalable	Difficult and slow to scale
Managed by provider	Managed by organization's IT team
Access from anywhere	Limited to physical location
Automatic updates	Manual upgrades required

Q: What are different components required in Cloud Computing?

- Client infrastructure (user devices – PC, mobile)
- Application (software/SaaS)
- Platform (PaaS – development environment)
- Storage (databases, object storage)
- Infrastructure (IaaS – virtual machines, networking)
- Management (orchestration, billing tools)
- Security (encryption, access control)
- Internet (connectivity layer)

Q: What is a Cloud Service?

Ans: A cloud service is any service made available to users on demand via the internet from a cloud computing provider's servers. It could be software, a platform, or infrastructure.

Q: List the basic clouds in Cloud Computing.

- Public Cloud (e.g., AWS, Azure, GCP)
- Private Cloud (e.g., VMware, OpenStack)
- Hybrid Cloud
- Community Cloud

Q: What are some issues with Cloud Computing?

- Security and privacy risks (data on third-party servers)
- Internet dependency (no internet = no access)
- Vendor lock-in (hard to switch providers)
- Downtime risk (even providers have outages)
- Limited control over infrastructure
- Compliance and legal issues (data stored in different countries)

Q: What are the different platforms of Cloud Architecture?

- Front-end platform (client-side: browser, app)
- Back-end platform (servers, storage, databases)
- Cloud-based delivery (IaaS, PaaS, SaaS)
- Network (internet connecting front-end and back-end)

Q: What are recent trends in Cloud Computing?

- Serverless computing (AWS Lambda)
- Edge computing (processing near the data source)
- AI/ML integration in cloud services
- Multi-cloud strategy
- Kubernetes and containerization (Docker)
- Cloud security improvements (Zero Trust model)

Q: Applications of Cloud Computing.

- Online storage and backup (Google Drive, Dropbox)
- Big data analysis
- Development and testing environments

- Email and communication (Gmail, Office 365)
- E-commerce platforms
- Healthcare data management
- Entertainment streaming (Netflix, YouTube)

■ [TRICKY] What is the difference between IaaS, PaaS, and SaaS? Who manages what?

Responsibility	IaaS	PaaS	SaaS
Hardware	Provider	Provider	Provider
OS	You	Provider	Provider
Runtime	You	Provider	Provider
Application	You	You	Provider
Data	You	You	Provider

■ [TRICKY] Is Google App Engine IaaS or PaaS? Why?

Ans: Google App Engine is PaaS. Because you only write and deploy your code — Google manages servers, OS, scaling, and runtime. You don't control the infrastructure.

■ [TRICKY] Can cloud computing work without the internet?

Ans: For public cloud — NO. But private cloud (on-premise) can work within a local network without internet. Edge computing also works locally. However, typical cloud services require internet connectivity.

SECTION 2: MICROSOFT AZURE

■ [IMPORTANT] What is Microsoft Azure?

Ans: Microsoft Azure is a cloud computing platform and infrastructure created by Microsoft for building, deploying, and managing applications and services through a global network of Microsoft-managed data centers. It provides IaaS, PaaS, and SaaS solutions.

■ [IMPORTANT] What are the services provided by Microsoft Azure?

- Azure Virtual Machines (IaaS – compute)
- Azure Blob Storage (object storage)
- Azure SQL Database (managed database)
- Azure App Service (host web apps – PaaS)
- Azure Active Directory (identity/access management)
- Azure Kubernetes Service (container orchestration)
- Azure CDN (content delivery network)
- Azure DevOps (CI/CD pipelines)

■ [IMPORTANT] What services does Windows Azure Operating System provide?

- Compute services (Virtual Machines, Cloud Services)
- Storage services (Blob, Queue, Table storage)
- Networking (Virtual Network, Load Balancer)
- Database services (SQL Azure)
- Identity services (Azure AD)
- Developer tools and SDKs

Q: What is Cloud Computing Architecture?

Ans: Cloud architecture has two main parts:

- Front-end: The user-facing side — client devices, browsers, applications used to access cloud
- Back-end: The provider side — servers, storage, databases, virtual machines, security systems

Ans: They communicate through the internet (middleware).

■ [IMPORTANT] What is Load Balancing in Cloud Computing?

Ans: Load balancing is the process of distributing incoming network traffic across multiple servers so no single server gets overloaded. It ensures high availability and reliability.

Ans: Example: Azure Load Balancer, AWS Elastic Load Balancer (ELB)

- Improves performance by distributing requests
- Provides fault tolerance (if one server fails, others handle traffic)
- Enables horizontal scaling

Q: What are large-scale cloud computing platforms?

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- IBM Cloud
- Oracle Cloud
- Alibaba Cloud

Q: What are large cloud providers and their databases?

Provider	Database Service
AWS	Amazon RDS, DynamoDB, Aurora
Microsoft Azure	Azure SQL, Cosmos DB
Google Cloud	Cloud SQL, Firestore, Bigtable
IBM Cloud	Db2 on Cloud, Cloudant

Q: What are open-source cloud computing platform databases?

- MySQL / PostgreSQL (relational)
- MongoDB (NoSQL document store)
- Cassandra (distributed NoSQL)
- CouchDB (document-oriented)
- Redis (in-memory key-value)
- OpenStack (open-source cloud platform)

■ [TRICKY] What is the difference between Azure Virtual Machine and Azure App Service?

Azure VM	Azure App Service
IaaS	PaaS
You manage OS, patches, runtime	Microsoft manages infrastructure
Full control	Limited control, easier to use
Good for custom configurations	Good for web apps, APIs quickly

■ [TRICKY] If Azure has an outage, whose fault is it and what can you do?

Ans: It is Microsoft's responsibility as per the SLA (Service Level Agreement). Azure guarantees 99.9–99.99% uptime. In case of outage: check Azure Status page, use multi-region deployment for redundancy, and claim SLA credits if uptime falls below guarantee.

SECTION 3: AMAZON EC2 (AWS)

■ [IMPORTANT] What is Amazon EC2?

Ans: EC2 stands for Elastic Compute Cloud. It is a web service provided by Amazon Web Services (AWS) that provides resizable virtual servers (instances) in the cloud. You can launch servers within minutes and pay only for what you use.

■ [IMPORTANT] What is an EC2 Instance?

Ans: An EC2 instance is a virtual machine running on Amazon's physical hardware. You choose: instance type (CPU/RAM size), operating system (via AMI), storage (EBS), and networking. It behaves exactly like a physical computer but runs in AWS data centers.

■ [IMPORTANT] What is AMI in EC2?

Ans: AMI stands for Amazon Machine Image. It is a pre-configured template/snapshot that contains:

- The operating system (Windows, Ubuntu, etc.)
- Pre-installed software/applications
- Configuration settings

Ans: AMI is used to launch EC2 instances. Think of AMI as a 'mold' and the EC2 instance as the 'object made from the mold'.

■ [IMPORTANT] How to launch an on-demand EC2 instance in AWS?

- Step 1: Log in to AWS Management Console
- Step 2: Go to EC2 Dashboard → Click 'Launch Instance'
- Step 3: Choose an AMI (e.g., Ubuntu 22.04 LTS)
- Step 4: Select instance type (e.g., t2.micro — free tier eligible)
- Step 5: Configure instance details (network, subnet, IAM role)
- Step 6: Add storage (EBS volume — default 8GB)
- Step 7: Add tags (Name = MyServer)
- Step 8: Configure Security Group (open port 22 for SSH, 80 for HTTP)
- Step 9: Create/Select Key Pair for SSH access
- Step 10: Click 'Launch Instance' — instance starts in ~1 minute

Q: What are the services provided by AWS?

- EC2 – Elastic Compute Cloud (Virtual Machines)
- S3 – Simple Storage Service (Object Storage)
- RDS – Relational Database Service
- Lambda – Serverless computing
- VPC – Virtual Private Cloud (networking)
- IAM – Identity and Access Management (security)
- CloudFront – CDN (Content Delivery Network)
- ELB – Elastic Load Balancer

■ [IMPORTANT] What is Virtualization?

Ans: Virtualization is the technology of creating a virtual (software-based) version of something physical — like a server, storage, or network. It allows multiple virtual machines (VMs) to run on one physical machine simultaneously, each with its own OS and applications.

- Type 1 Hypervisor (Bare Metal): Runs directly on hardware — KVM, VMware ESXi, Hyper-V
- Type 2 Hypervisor (Hosted): Runs on top of OS — VirtualBox, VMware Workstation

■ [IMPORTANT] What is a Hypervisor in Cloud Computing?

Ans: A hypervisor is software that creates and manages virtual machines. It acts as a layer between the physical hardware and the VMs. It allocates CPU, RAM, and storage to each VM.

Type 1 (Bare Metal)	Type 2 (Hosted)
Runs directly on hardware	Runs on top of an OS

Type 1 (Bare Metal)	Type 2 (Hosted)
Better performance	Slightly slower (extra OS layer)
Used in production/cloud	Used for testing/development
KVM, VMware ESXi, Hyper-V	VirtualBox, VMware Workstation

■ [TRICKY] What is the difference between On-Demand, Reserved, and Spot instances in EC2?

Type	Description	Cost
On-Demand	Pay per hour, no commitment	Highest
Reserved	1-3 year commitment, predictable workload	Up to 75% cheaper
Spot	Use unused AWS capacity, can be terminated at any time	Up to 90% cheaper

■ [TRICKY] What is EBS in AWS EC2?

Ans: EBS stands for Elastic Block Store. It is a persistent storage volume attached to an EC2 instance — like a hard drive for your virtual machine. Unlike instance storage, EBS data persists even after the instance is stopped or terminated.

■ [TRICKY] If you stop an EC2 instance, do you still get charged?

Ans: Partially. When an EC2 instance is STOPPED: you are NOT charged for compute (CPU/RAM), BUT you ARE still charged for EBS storage attached to it. To stop all charges, you must TERMINATE the instance.

SECTION 4: GOOGLE APP ENGINE

■ [IMPORTANT] What is Google App Engine?

Ans: Google App Engine (GAE) is a Platform as a Service (PaaS) offered by Google Cloud. It allows developers to build and host web applications and mobile backends on Google's managed infrastructure. You only write code — Google handles servers, load balancing, and scaling automatically.

■ [IMPORTANT] What are the advantages of Google App Engine?

- No server management – Google handles all infrastructure
- Automatic scaling – Scales up/down based on traffic
- Multi-language support – Python, Java, PHP, Go, Node.js
- Built-in security and monitoring
- Integrated with other Google services (Cloud SQL, Datastore)
- Free tier available for small applications
- Version management – deploy multiple versions, easy rollback

■ [IMPORTANT] What are the steps to install and configure Google App Engine?

- Step 1: Create a Google Cloud account at console.cloud.google.com
- Step 2: Create a new project in Google Cloud Console
- Step 3: Install Google Cloud SDK (gcloud CLI)
- Step 4: Run 'gcloud init' to authenticate and set default project
- Step 5: Install App Engine extension: `gcloud components install app-engine-python`
- Step 6: Create your application code (e.g., `main.py` for Python)
- Step 7: Create `app.yaml` configuration file (defines runtime, handlers)
- Step 8: Test locally using '`dev_appserver.py app.yaml`'
- Step 9: Deploy using '`gcloud app deploy`'
- Step 10: View app using '`gcloud app browse`'

Q: What does `app.yaml` contain?

Ans: `app.yaml` is the configuration file for Google App Engine. It contains:

- `runtime`: `python39` (language and version)
- `handlers`: URL routing rules
- `env_variables`: environment variables
- `automatic_scaling` settings
- `instance_class` (machine type)

■ [TRICKY] What is the difference between Standard Environment and Flexible Environment in App Engine?

Standard Environment	Flexible Environment
Runs in sandbox	Runs in Docker containers on VMs
Scales to zero (free when idle)	Minimum 1 instance always running
Limited language support	Any language via Docker
Faster to scale	More flexibility and customization
Cheaper for low traffic	Better for sustained traffic

■ [TRICKY] Google App Engine vs AWS EC2 — which is better?

Ans: Neither is universally 'better' — it depends on the use case:

App Engine (PaaS)	AWS EC2 (IaaS)
No server management	Full control over server
Less flexible	Highly flexible

App Engine (PaaS)	AWS EC2 (IaaS)
Best for web apps	Best for complex custom environments
Auto-scaling built-in	Manual or auto-scaling (configure yourself)

SECTION 5: VIRTUALIZATION USING KVM

■ [IMPORTANT] What is KVM?

Ans: KVM stands for Kernel-based Virtual Machine. It is an open-source virtualization technology built directly into the Linux kernel (since Linux 2.6.20). KVM converts the Linux kernel into a Type 1 (bare-metal) hypervisor, allowing you to run multiple virtual machines on a single Linux host.

■ [IMPORTANT] How to install and configure KVM on Ubuntu?

- Step 1: Check if CPU supports virtualization: `egrep -c '(vmx|svm)' /proc/cpuinfo` (output > 0 means supported)
- Step 2: Install KVM packages: `sudo apt update && sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager`
- Step 3: Add user to groups: `sudo adduser $USER kvm && sudo adduser $USER libvirt`
- Step 4: Start and enable libvirt service: `sudo systemctl start libvirtd && sudo systemctl enable libvirtd`
- Step 5: Verify installation: `virsh list --all`
- Step 6: Open Virtual Machine Manager (GUI): `virt-manager`
- Step 7: Create new VM: Click 'New', choose ISO, allocate RAM and disk, complete setup

Q: What is the difference between KVM and VirtualBox?

KVM	VirtualBox
Type 1 Hypervisor (built into kernel)	Type 2 Hypervisor (runs on OS)
Linux only (host)	Cross-platform (Windows, Mac, Linux)
Better performance	Slightly slower
Used in production/server environments	Used for development/testing
Command-line + virt-manager GUI	User-friendly GUI

■ [TRICKY] What is QEMU? How does it relate to KVM?

Ans: QEMU (Quick Emulator) is an open-source emulator that simulates hardware. KVM uses QEMU as its userspace component. Together: KVM handles CPU/memory virtualization in the kernel, while QEMU emulates I/O devices (disk, network). They work as a team — you see them as 'QEMU-KVM'.

■ [TRICKY] What is libvirt?

Ans: Libvirt is an open-source API and management tool for managing virtualization platforms including KVM. It provides a unified interface to manage VMs using commands (`virsh`) or GUI (`virt-manager`). Common `virsh` commands:

- `virsh list --all` (list all VMs)
- `virsh start vmname` (start a VM)
- `virsh shutdown vmname` (gracefully shut down)
- `virsh destroy vmname` (force stop)
- `virsh snapshot-create vmname` (take snapshot)

SECTION 6: SALESFORCE.COM & APEX PROGRAMMING

■ [IMPORTANT] What is Salesforce.com?

Ans: Salesforce.com Inc. is an American cloud-based software company. It provides a CRM (Customer Relationship Management) platform that helps businesses manage customers, sales, marketing, and support — entirely on the cloud. It was founded in 1999 by Marc Benioff.

Q: What is Lightning Platform (Force.com)?

Ans: Lightning Platform (formerly Force.com) is Salesforce's PaaS offering that allows developers to build custom business applications on top of Salesforce infrastructure. It includes: Apex (programming language), Visualforce/Lightning Web Components (UI framework), SOQL (database query language), and Workflow automation tools.

■ [IMPORTANT] What is Apex?

Ans: Apex is a strongly-typed, object-oriented programming language developed by Salesforce. It is similar to Java in syntax and runs entirely on the Salesforce cloud platform (Force.com). It is used to write business logic that runs on Salesforce servers.

■ [IMPORTANT] What are the features of Apex?

- Strongly typed – Variables must have declared data types
- Object-oriented – Supports classes, objects, inheritance, interfaces
- Integrated with database – Uses SOQL (Salesforce Object Query Language) directly in code
- Multi-tenant aware – Runs in Salesforce's shared environment with governor limits
- Supports triggers – Auto-execute code when records are inserted/updated/deleted
- Supports batch processing – Handle millions of records asynchronously
- Built-in testing framework – @isTest annotation for unit tests
- Web service support – Create REST and SOAP APIs

Q: What are the applications of Apex?

- Writing triggers (auto-run code when data changes in Salesforce)
- Creating Visualforce page controllers (server-side logic for UI)
- Batch Apex (process large datasets in chunks)
- Scheduled Apex (run code at specific times, like a cron job)
- Building REST/SOAP web services
- Callouts (calling external APIs from Salesforce)
- Custom validation logic

Q: List some Apex Code Development Tools.

- Salesforce Developer Console – Browser-based IDE within Salesforce
- VS Code with Salesforce Extension Pack (SFDX) – Most popular modern tool
- MavensMate – Eclipse/Sublime Text plugin (older)
- Force.com IDE – Eclipse plugin (legacy)
- Workbench – Web-based tool for SOQL queries and metadata

■ [IMPORTANT] What are the steps to create an application using Apex Programming Language?

- Step 1: Create a free Salesforce Developer account at developer.salesforce.com
- Step 2: Log in and go to Setup
- Step 3: Open Developer Console (Gear icon → Developer Console)
- Step 4: Go to File → New → Apex Class
- Step 5: Write your Apex class with business logic
- Step 6: Save and compile (errors shown immediately)
- Step 7: Write test class with @isTest annotation (required for deployment)
- Step 8: Run tests via Test → Run All
- Step 9: Use the class in Triggers, Visualforce, or Lightning Components

■ **[IMPORTANT] What is the difference between Custom Application and Console Application in Salesforce?**

Custom Application	Console Application
Standard tab-based navigation	Multi-tab workspace interface
One record viewed at a time	Multiple records open simultaneously
Used for general business users	Used for support/call center agents
Simpler interface	Complex, power-user interface
Created via App Manager	Requires Service Cloud Console license

■ **[IMPORTANT] How to create a Custom Application using Salesforce Classic?**

- Step 1: Log in to Salesforce → Go to Setup
- Step 2: Navigate to Build → Create → Apps
- Step 3: Click 'New' to create a new app
- Step 4: Choose 'Custom app' and click Next
- Step 5: Enter App Label (name), App Name, and optionally upload a logo
- Step 6: Choose navigation tabs to include (standard + custom objects)
- Step 7: Assign the app to user profiles (who can see this app)
- Step 8: Click Save — app appears in the app launcher

■ **[IMPORTANT] What are the steps to create a custom application using Salesforce (Mini Project)?**

- Step 1: Log in to Salesforce Developer Org
- Step 2: Go to Setup → App Manager → New Lightning App
- Step 3: Give App Name, Description, choose branding
- Step 4: Select App Options (navigation type: standard or console)
- Step 5: Add utility items if needed
- Step 6: Add Navigation Items (Home, Contacts, Custom Objects, etc.)
- Step 7: Assign User Profiles who can access this app
- Step 8: Save and Finish
- Step 9: Create custom objects/fields if required via Object Manager
- Step 10: Build the app logic using Apex, Flows, or Process Builder

■ **[TRICKY] What is a Trigger in Salesforce Apex? When does it fire?**

Ans: A trigger is Apex code that executes automatically before or after data manipulation events on Salesforce objects (records).

Ans: Trigger Events:

- before insert / after insert
- before update / after update
- before delete / after delete
- after undelete

Ans: Example: An 'after insert' trigger on Contact can automatically send a welcome email whenever a new contact is created.

■ **[TRICKY] What are Governor Limits in Salesforce Apex?**

Ans: Governor limits are runtime limits enforced by Salesforce to ensure no single tenant uses too many shared resources (since Salesforce is multi-tenant). Examples:

- Max 100 SOQL queries per transaction
- Max 150 DML statements per transaction
- Max 50,000 records returned by SOQL
- Max CPU time: 10,000ms

Note: Governor limits are a very common tricky exam question — remember the number 100 SOQL queries!

■ [TRICKY] What is SOQL? How is it different from SQL?

SOQL	SQL
Salesforce Object Query Language	Structured Query Language
Only SELECT (no INSERT/UPDATE/DELETE)	Full DML support
Used within Apex code	Used in relational databases
Works only on Salesforce objects	Works on any relational database
Example: SELECT Id, Name FROM Account	SELECT * FROM Account

■ [TRICKY] What is the difference between Apex Class and Apex Trigger?

Apex Class	Apex Trigger
Reusable block of code with methods	Code that fires on database events
Called explicitly (by other code, UI, API)	Called automatically by Salesforce
Can have constructors, interfaces	No constructors
Example: Helper class, Controller	Example: ContactTrigger on Contact

SECTION 7: ADDITIONAL TRICKY & IMPORTANT QUESTIONS

■ [TRICKY] What is the difference between Scalability and Elasticity in Cloud?

Scalability	Elasticity
Ability to handle growing workload	Ability to automatically scale up AND down
Usually planned/manual	Automated and dynamic
Long-term capacity planning	Short-term fluctuation handling
Example: Adding servers for a growing business	Example: Auto-scale during Black Friday sale, scale down after

■ [TRICKY] What is Multi-tenancy in Cloud Computing?

Ans: Multi-tenancy means multiple customers (tenants) share the same physical infrastructure and software platform, but their data is isolated from each other. Salesforce is a classic example — thousands of companies use the same servers but cannot see each other's data.

Ans: Benefits: Cost-sharing, resource efficiency. **Risk:** 'Noisy neighbor' effect (one tenant using too many resources affects others).

■ [TRICKY] What is the difference between Public IP and Elastic IP in AWS?

Public IP	Elastic IP
Changes every time instance restarts	Static — remains the same permanently
Assigned automatically by AWS	You request and assign it manually
Free	Free only when attached to running instance
Cannot be reassigned to another instance	Can be remapped to any instance in your account

■ [TRICKY] What is Serverless Computing? Is it really 'no server'?

Ans: Serverless computing means the developer does NOT manage servers — but servers DO still exist. The cloud provider automatically provisions, scales, and manages the servers. You only write functions/code and pay per execution.

Ans: Examples: AWS Lambda, Google Cloud Functions, Azure Functions

Note: In an exam: 'Serverless' means no server MANAGEMENT, not no servers physically.

■ [TRICKY] What is Edge Computing? How is it different from Cloud Computing?

Cloud Computing	Edge Computing
Processing in centralized data centers	Processing near the data source (at the 'edge')
Higher latency (data travels far)	Lower latency (data processed locally)
Requires internet connectivity	Can work without constant internet
Example: Netflix server streaming	Example: Autonomous car processing sensor data locally

■ [IMPORTANT] What is SLA in Cloud Computing?

Ans: SLA stands for Service Level Agreement. It is a formal contract between the cloud provider and the customer that defines the guaranteed level of service — especially uptime/availability.

- AWS EC2 SLA: 99.99% uptime
- Azure VM SLA: 99.9-99.99% uptime
- 99.9% uptime = ~8.7 hours downtime per year
- 99.99% uptime = ~52 minutes downtime per year

■ [TRICKY] What is the difference between Fault Tolerance and High Availability?

Fault Tolerance	High Availability
System continues working EVEN IF components fail	System is designed to minimize downtime
Zero downtime goal	Minimal/brief downtime acceptable
More expensive (full redundancy)	Less expensive
Example: RAID storage, redundant power supplies	Example: Load balancer + multiple servers

Q: What is CDN (Content Delivery Network)?

Ans: A CDN is a network of geographically distributed servers that cache and deliver web content to users from the nearest server — reducing load times. Example: If a user in India requests a website hosted in the USA, the CDN server in India serves the cached content instead.

Ans: Cloud CDN examples: AWS CloudFront, Azure CDN, Google Cloud CDN

Q: What is Docker? How does it relate to Cloud?

Ans: Docker is a containerization platform that packages applications and all their dependencies into lightweight, portable containers. Unlike VMs, containers share the host OS kernel (no separate OS per container), making them faster and more efficient. Cloud platforms (AWS, Azure, GCP) all support Docker containers natively.

■ [TRICKY] What is the difference between a Container and a Virtual Machine?

Container (Docker)	Virtual Machine
Shares host OS kernel	Has its own full OS
Lightweight (MBs)	Heavy (GBs)
Starts in seconds	Takes minutes to boot
Less isolation	Strong isolation
Used for microservices/apps	Used for full OS environments

QUICK REVISION CHEAT SHEET

Topic	Key Point to Remember
AMI	Amazon Machine Image = blueprint/template to launch EC2 instances
EC2	Elastic Compute Cloud = virtual machine on AWS
GAE	Google App Engine = PaaS, auto-scales, no server management
KVM	Kernel-based Virtual Machine = Type 1 hypervisor in Linux kernel
Apex	Salesforce's OOP language, similar to Java, runs on Force.com
SOQL	SELECT only, for querying Salesforce objects, used inside Apex
Trigger	Apex code that fires automatically on Salesforce DML events
IaaS/PaaS/SaaS	EC2=IaaS, App Engine=PaaS, Salesforce/Gmail=SaaS
Hypervisor	Type1=bare metal(KVM), Type2=hosted(VirtualBox)
Load Balancing	Distributes traffic across multiple servers
Elasticity	Auto scale UP and DOWN based on demand
Multi-tenancy	Multiple users share same infra, data isolated
Governor Limits	Salesforce limits: 100 SOQL queries per transaction
Vertical Scaling	Add more power to same machine (more RAM/CPU)
Horizontal Scaling	Add more machines to distribute load

Best of Luck for your exam tomorrow! You've got this! ■

Savitribai Phule Pune University | Laboratory Practice-II (Cloud Computing)